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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

AC2101-1- COMMUNICATION SYSTEMS

(For AC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In an AM system, the carrier signal has an amplitude of 10 V, and the modulating signal has a peak amplitude of 4 V. What is the modulation index?
A) 0.4 B) 0.6
C) 0.8 D) 1.2
2. A square law modulator is used for AM generation. If the carrier frequency is 1 MHz and the modulating frequency is 5 kHz, what will be the frequencies in the modulated signal?
A) 1 MHz and 1.005 MHz B) 1 MHz, 995 kHz, and 1.005 MHz
C) 995 kHz and 1.005 MHz D) 5 kHz and 1 MHz
3. A 1 kHz sine wave is modulated using SSB on a 100 kHz carrier. What will be the bandwidth of the modulated signal?
A) 1 kHz B) 2 kHz
C) 100 kHz D) 200 kHz
4. In Double-Sideband Suppressed Carrier (DSB-SC) modulation, the carrier:
A) Is transmitted along with the sidebands B) Is suppressed to save power
C) Contains all the information D) Is amplified
5. Which device is commonly used for FM generation?
A) Multiplier B) Phase detector
C) Voltage-controlled oscillator D) Envelope detector
6. The equation of amplitude wave is given by $s(t)=20[1+0.8\cos(2\pi\times 10^3t)]\cos(4\pi\times 10^5t)$, What is the total power of the AM wave?
A) 200 W B) 264 W
C) 300 W D) 350 W
7. The FM signal is given by $s(t)=10\cos(2\pi\times 10^6t+20\sin(2\pi\times 10^3t))$, What is the approximate bandwidth of the signal according to Carson's rule?
A) 2 kHz B) 20 kHz
C) 82 kHz D) 42 kHz
8. Which of the following is a key characteristic of white noise?
A) It has a finite power spectral density B) Its power spectral density is constant over all frequencies
C) It occurs only at high frequencies D) It is found only in audio systems
9. What is noise equivalent bandwidth (NEB)?
A) Bandwidth required for maximum noise generation B) Bandwidth of an equivalent rectangular filter that passes the same noise power
C) Maximum possible filter bandwidth D) Bandwidth when no noise is present

10. What is the Friis formula for the noise figure of a cascaded system?
 A) $F_{\text{total}} = F_1 + (F_2/G_1) + (F_3/(G_1G_2))$ B) $F_{\text{total}} = G_1 \times F_1$
 C) $F_{\text{total}} = F_1 - F_2 + G_1$ D) $F_{\text{total}} = F_1 + F_2 + G_2$
11. The equivalent noise temperature is defined as:
 A) A temperature where the gain of the amplifier is zero B) The temperature at which the noise power equals that produced by a perfect resistor
 C) Temperature when signal power is constant D) Temperature where thermal noise vanishes
12. Flat-top sampling is preferred over natural sampling because:
 A) It provides better fidelity B) It allows easier signal reconstruction
 C) It eliminates aliasing completely D) It minimizes quantization noise
13. Which type of quantizer provides better performance for low amplitude signals?
 A) Uniform quantizer B) Linear quantizer
 C) Adaptive quantizer D) Non-uniform quantizer
14. A-law and μ -law companding techniques are used to:
 A) Reduce sampling rate B) Decrease transmission bandwidth
 C) Improve signal-to-noise ratio for low amplitude signals D) Eliminate quantization noise
15. In an 8-bit PCM system, the number of quantization levels is:
 A) 128 B) 512
 C) 256 D) 1024
16. A signal is quantized into 16 levels. If the dynamic range of the signal is 10 volts, what is the step size for the quantizer?
 A) 0.5 V B) 0.625 V
 C) 1.0 V D) 2.5 V
17. In Delta Modulation (DM), which factor primarily determines the step size?
 A) Quantization level B) Sampling frequency
 C) Modulation index D) Slope of the input signal
18. DPSK eliminates the need for:
 A) Carrier synchronization B) Signal bandwidth
 C) Noise reduction D) Phase modulation
19. Quadrature Amplitude Modulation (QAM) utilizes which components as the two signal dimensions?
 A) Only In-phase (I) component B) Only Quadrature (Q) component
 C) Both In-phase (I) and Quadrature (Q) components D) Neither In-phase (I) nor Quadrature (Q) components
20. How does QPSK compare to BPSK in terms of bandwidth efficiency?
 A) QPSK requires twice the bandwidth B) QPSK requires half the bandwidth
 C) Both require the same bandwidth D) BPSK is more efficient than QPSK

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Illustrate the working of a square law modulator for AM generation with a neat block diagram and waveforms.	8	L*2	1	1,2
	b) Define frequency modulation. Derive the mathematical expression for a single tone frequency modulation and draw the necessary waveforms.	8	L3	2	1,2
2.	a) Discuss the advantages and disadvantages of Frequency Modulation (FM) compared to Amplitude Modulation (AM) in terms of noise immunity, signal quality, bandwidth efficiency, and power requirements.	5	L2	1	1,2
	b) Compare and contrast Narrowband frequency Modulated (NBFM) signal and Wideband Frequency Modulated (WBFM) signal on various parameters.	5	L2	2	1,2

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|-------|---|---|----|---|-----|
| c) | Explain the balanced frequency discrimination method of FM detection with suitable block diagram/ circuit diagrams. | 6 | L2 | 2 | 1,2 |
| 3. a) | Explain the generation of DSB-SC modulated wave using a ring modulator with relevant circuit diagram and waveforms. | 8 | L2 | 1 | 1,2 |
| b) | Explain the direct method of Frequency Modulation (FM) generation using a Voltage-Controlled Oscillator (VCO). Illustrate with a relevant block diagram and derive the necessary expressions. | 8 | L3 | 2 | 1,2 |

Unit – II

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|-------|--|---|----|---|-----|
| 4. a) | List and explain different types of noise in communication systems. Discuss the impact of each type on the performance of communication systems. | 8 | L2 | 3 | 1,2 |
| b) | Illustrate the process of quantization and quantization errors by drawing a quantized signal along with its corresponding error curve. | 8 | L3 | 4 | 1,2 |
| 5. a) | Explain the concept of thermal noise. Derive the formula for thermal noise power. | 5 | L2 | 3 | 1,2 |
| b) | Calculate the RMS noise voltage and thermal noise power appearing across a 20 kΩ resistor at 25°C temperature with an effective noise bandwidth of 10 kHz. | 4 | L2 | 3 | 1,2 |
| c) | Explain natural sampling and flat-top sampling, compare their characteristics, and discuss their advantages and limitations. | 7 | L2 | 4 | 1,2 |
| 6. a) | Explain the concepts of Noise Factor and Noise Figure. Derive the expression for output noise power in terms of the Noise Figure (F) and Gain (G) of the system. | 6 | L3 | 3 | 1,2 |
| b) | Describe the basic elements of a Pulse Code Modulation (PCM) system with a block diagram. | 6 | L2 | 4 | 1,2 |
| c) | A signal $x(t)=5\cos(2000\pi t)+3\cos(6000\pi t)$ is sampled at 5000 samples per second. | | | | |
| i) | Determine if the sampling rate satisfies the Nyquist criterion. | | | | |
| ii) | Find the minimum sampling rate required to avoid aliasing. | 4 | L2 | 4 | 1,2 |

Unit – III

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| 7. a) | Discuss the working principle of Delta Modulation (DM). Illustrate the block diagrams of the Delta Modulation transmitter and receiver, detailing the function of each block. | 8 | L2 | 5 | 1,2 |
| b) | Explain the working of QPSK modulation and demodulation, with the necessary block diagrams and waveforms. | 8 | L2 | 5 | 1,2 |
| 8. a) | Illustrate the generation of three basic forms of signaling binary information: ASK, FSK and PSK with suitable block diagrams and waveforms. | 8 | L2 | 5 | 1,2 |
| b) | Explain the working principle of Differential Phase Shift Keying (DPSK). Draw and describe the block diagram for DPSK demodulation. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
